

The recombination landscape of introgression in yeast

Reviewed Preprint

v2 • October 16, 2024

Revised by authors

Reviewed Preprint

v1 • March 15, 2024

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Abstract

Meiotic recombination is an evolutionary force that acts by breaking up genomic linkage, increasing the efficacy of selection. Recombination is initiated with a double-strand break which is resolved via a crossover, which involves the reciprocal exchange of genetic material between homologous chromosomes, or a non-crossover, which results in small tracts of non-reciprocal exchange of genetic material. Crossover and non-crossover rates vary between species, populations, individuals, and across the genome. In recent years, recombination rate has been associated with the distribution of ancestry derived from past interspecific hybridization (introgression) in a variety of species. We explore this interaction of recombination and introgression by sequencing spores and detecting crossovers and non-crossovers from two crosses of the yeast *Saccharomyces uvarum*. One cross is between strains which each contain introgression from their sister species, *S. eubayanus*, while the other cross has no introgression present. We find that the recombination landscape is significantly different between *S. uvarum* crosses, and that some of these differences can be explained by the presence of introgression in one cross. Crossovers are reduced and non-crossovers are increased in heterozygous introgression compared to syntenic regions in the cross without introgression. This translates to reduced allele shuffling within introgressed regions, and an overall reduction of shuffling on most chromosomes with introgression compared to the syntenic regions and chromosomes without introgression. Our results suggest that hybridization can significantly influence the recombination landscape, and that the reduction in allele shuffling contributes to the initial purging of introgression in the generations following a hybridization event.

eLife Assessment

This study aims to analyse the effect of polymorphism on meiotic recombination in subspecies of *Saccharomyces*. The detection of reciprocal and non-reciprocal events is based on sequencing the haploid products of meiosis, and frequencies are compared between strains having introgressed genomic segments and strains lacking such segments. Unfortunately, the method used are **inadequate** for quantifying the non-reciprocal events.

<https://doi.org/10.7554/eLife.96184.2.sa3>

Introduction

Recombination is the exchange of genetic material between homologous chromosomes during meiosis and is a staple of eukaryotic sexual reproduction. While the processes involved in recombination are largely conserved (Arter & Keeney, 2023 [↗](#)), recombination rates vary between sexes, populations, and species (Smukowski & Noor, 2011; Stapley et al., 2017 [↗](#)).

Recombination rates also vary along the genome, with conflicting patterns of enriched or depleted recombination in promoter regions and punctate or dispersed recombination depending on the species (Auton et al., 2013 [↗](#); Rockman & Kruglyak, 2009 [↗](#); Singhal et al., 2015 [↗](#); Smukowski & Heil et al., 2015 [↗](#)). These patterns in recombination can affect pairing of alleles after meiosis—in other words, the shuffling of alleles—in a population. Much of the evolutionary advantage of recombination is understood to originate from its role in shuffling alleles, which increases the number of different allele combinations segregating in a population. The increase in allele combinations can reduce selection interference—the effect that genetically linked sites have on the evolutionary fate of either beneficial or deleterious alleles (Felsenstein, 1974 [↗](#); Hill & Robertson, 1966 [↗](#); McDonald et al., 2016 [↗](#); McGaugh et al., 2012 [↗](#)).

How much allele decoupling is produced by recombination will depend on the type of recombination event. Each recombination event begins with the severing of both strands of a sister chromatid of one of the homologous chromosomes in what is referred to as a meiotic double-strand break (DSB) (Keeney, 2001 [↗](#)). A meiotic DSB is repaired as a crossover (CO), which results in the reciprocal exchange of genetic information between homologous chromosomes, or as a non-crossover (NCO) gene conversion—where a small segment (typically 100–2000 bp) of a chromosome is replaced by a copy of its homolog (Chovnick et al., 1971 [↗](#); Hilliker et al., 1994 [↗](#); Jeffreys & May, 2004 [↗](#); Judd & Petes, 1988 [↗](#)). COs generally produce more allele shuffling, and therefore degrade linkage faster than NCOs. However, NCOs can occur in regions where COs are typically suppressed, like centromeres and inversions (Korunes & Noor, 2019 [↗](#); Mancera et al., 2008 [↗](#); Miller et al., 2016 [↗](#); Schaeffer & Anderson, 2005 [↗](#); Shi et al., 2010 [↗](#); Talbert & Henikoff, 2010 [↗](#); Wijnker et al., 2013 [↗](#)). NCOs are also crucial to reducing linkage within coding regions and, unlike COs, result in 3:1 allele ratio in the meiotic product at heterozygous sites, potentially changing allele frequencies (Korunes & Noor, 2017 [↗](#)).

Variation in the number and distribution of COs and NCOs, and their respective associated effects on linkage, have important implications for molecular evolution. Recombination has long been appreciated to play a role in the distribution of various genomic features including nucleotide diversity. Nucleotide diversity has a positive correlation with recombination rate in a number of species, interpreted to result from selective sweeps and background selection removing genetic variation in regions of low recombination (Begun & Aquadro, 1992 [↗](#); Charlesworth et al., 1993 [↗](#); Smith & Haigh, 1974 [↗](#)). Similarly, recombination breaking up genetic associations is particularly notable in the context of interspecific hybridization. In first-generation (F_1) hybrids, the hybrid genome is heterozygous for each parental species. If the hybrids then back-cross to one of the parental species, recombination will produce genomes that are a mosaic of genetic information from the two species with a minor contribution from the species that was not backcrossed to (introgression) (Aguillon et al., 2022 [↗](#)). When each population has evolved alleles that are deleterious when present in the background of the other population (the Dobzhansky-Muller hybrid incompatibility model) we expect introgressed regions with low rates of recombination to be quickly purged from the population, as the accumulation of incompatible alleles incurs a steep fitness cost. In contrast, when introgressed regions have high recombination rates, the break up of genetic associations will reduce selective interference between the incompatible alleles and their surrounding haplotypes, allowing for neutral and beneficial alleles brought in with the introgression to escape the fate of neighboring incompatibilities (Barton & Bengtsson, 1986 [↗](#);

Butlin, 2005 [↗](#); Moran et al., 2021 [↗](#); Nachman, M. & Payseur, B., 2012; Schumer et al., 2018 [↗](#); Veller et al., 2023 [↗](#)). This theory is supported empirically through enrichment of introgressed segments in regions of higher recombination in a number of organisms including *Mimulus*, maize, butterflies, swordtail fish, stickleback, and humans (Brandvain et al., 2014 [↗](#); Calfee et al., 2021 [↗](#); Edelman et al., 2019 [↗](#); Martin et al., 2019 [↗](#); Ravinet et al., 2018 [↗](#); Schumer et al., 2018 [↗](#)).

This positive correlation between introgressed ancestry and recombination is emerging as a nearly ubiquitous pattern (though see (Dagilis & Matute, 2023 [↗](#); Duranton & Pool, 2022 [↗](#); Pool, 2015 [↗](#))), however, it is unclear how these observations relate to the known effect of sequence divergence on DSB repair. Introgression, particularly between highly diverged species, can have low sequence similarity with the genomic region it is replacing. A DSB in a region of low sequence similarity will recruit mismatch repair proteins, which ensure COs are occurring between homologous chromosomes and at equivalent positions to prevent ectopic recombination (Harfe & Jinks-Robertson, 2000 [↗](#); Hunter et al., 1996 [↗](#)). Mismatch repair proteins typically reduce the frequency of CO events as sequence divergence increases (Borts & Haber, 1987 [↗](#); Chen & Jinks-Robertson, 1999 [↗](#); Cooper et al., 2021 [↗](#); Martini et al., 2011 [↗](#)). Given that heterozygous introgression will have divergent sequences, we expect a decrease in COs, and possibly an increase in NCOs as DSBs fail to be repaired as COs in heterozygous introgression.

To help us understand this interaction of introgression and recombination, and identify patterns in CO and NCO in closely related populations, we utilized the budding yeast *Saccharomyces*. Yeasts provide an excellent opportunity to study DSB repair, as we can readily isolate and collect all four meiotic products of a given meiosis and detect both CO and more elusive NCO events (**Figure 1A** [↗](#)) (Brion et al., 2017 [↗](#); Gerton et al., 2000 [↗](#); Liu et al., 2018 [↗](#), 2019 [↗](#); Mancera et al., 2008 [↗](#)). Recombination rates vary between strains of *S. cerevisiae* (Cubillos et al., 2011 [↗](#); Raffoux et al., 2018 [↗](#)) and between *S. cerevisiae* and its sister species *S. paradoxus* (Liu et al., 2019 [↗](#); Tsai et al., 2010 [↗](#)). Strains of different *Saccharomyces* species have often hybridized with other species and carry introgressed DNA from these events (Albertin et al., 2018 [↗](#); Almeida et al., 2014 [↗](#); Bendixsen et al., 2022 [↗](#); D'Angiolo et al., 2020; Langdon et al., 2019 [↗](#); Stelkens & Bendixsen, 2022 [↗](#); Tellini et al., 2023 [↗](#)).

In this study, we look at patterns of recombination and introgression by crossing two pairs of Holarctic *Saccharomyces uvarum* strains. One pair of strains was isolated from natural environments in North America and the other pair was isolated from European fermentation environments (Almeida et al., 2014 [↗](#)). The *S. uvarum* strains isolated from European fermentation environments each carry introgression from their sister species, *Saccharomyces eubayanus*, which is approximately 6% divergent from *S. uvarum* (Almeida et al., 2014 [↗](#); Langdon et al., 2020 [↗](#); Nespolo et al., 2020 [↗](#)). The diploid F₁ genome of these strains is heterozygous for nine different introgressions which make up approximately 10% of the genome (**Figure 1B** [↗](#)). The strains from the North American cross do not carry *S. eubayanus* introgression, thus allowing us to assess the impact of introgression on the recombination landscape. We obtained whole genome sequencing data from individual meiotic events from the first offspring generation of each cross and used this data to detect CO and NCO events along the genome (**Figure 1A** [↗](#)). From these maps, we aim to understand (i) how patterns of CO and NCO differ between closely related strains, (ii) how regions of introgression differ in their CO and NCO patterns, and (iii) how these different patterns affect shuffling of alleles locally and at the chromosome level. Understanding these objectives will provide us novel insights into how introgression impacts the recombination landscape.

Results

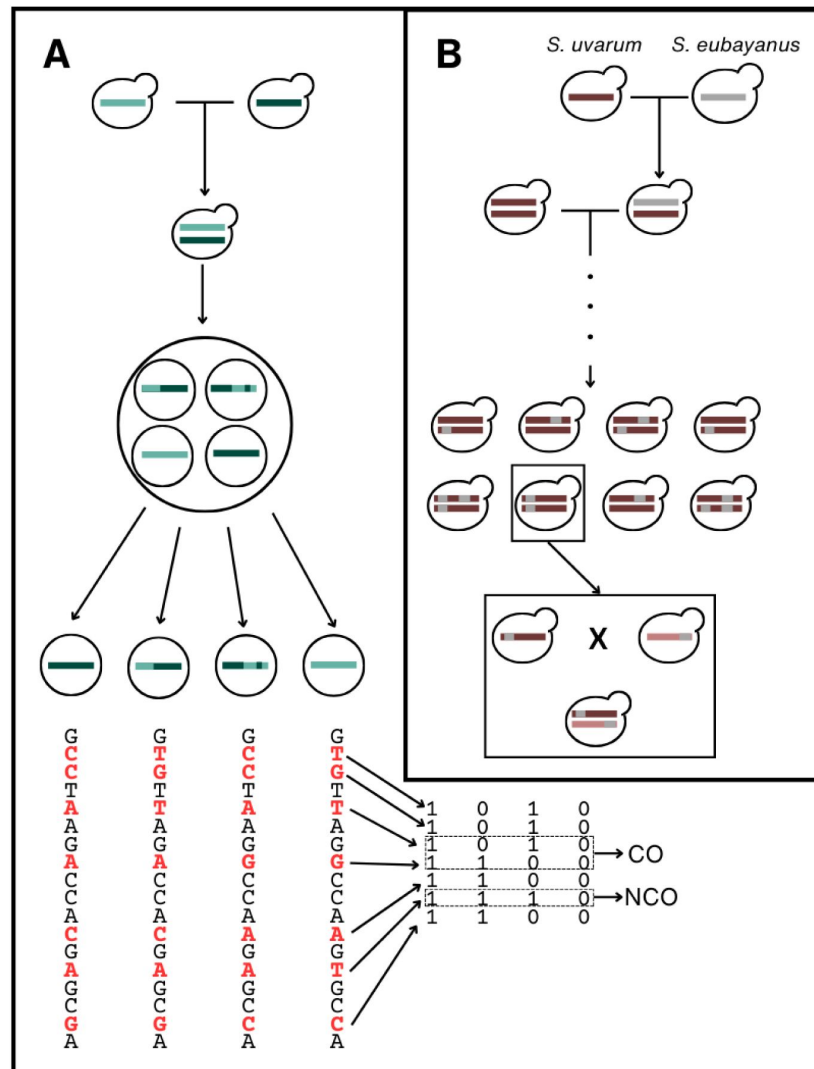


Figure 1

(A) Visual representation of how our crosses were conducted. Haploid yeast from two genetically distinct parental strains are mated to produce a heterozygous diploid. Meiosis is induced and the resulting meiotic products (tetrads) are manually dissected, and each haploid meiotic product is grown mitotically to obtain enough material for DNA extraction and whole genome sequencing. We then call SNPs on the resulting sequences and retain loci with fixed differences between parents. These loci are then coded as 1 or 0 depending on the parent of origin and the CrossOver software detects COs and NCOs. (B) A schematic of how introgression likely arose in the strains sampled from fermentation environments. These introgressions are likely due to *S. eubayanus* hybridizing with *S. uvarum* at some point in the past, resulting in F1 hybrids that then potentially crossed with other *S. uvarum* individuals for some number of generations. Eventually, the *S. eubayanus* ancestry was degraded in the population of *S. uvarum* until the introgressions we observe today remained, potentially segregating in the population. A similar process likely happened in each of the parental strains we utilized, but with different introgressions remaining in each strain. We crossed haploid individuals from two parental strains, resulting in a diploid that is heterozygous for each introgression.

The recombination landscape differs dramatically between closely related crosses

We isolated and sequenced products of 48 meioses (192 haploid spores) for two crosses of *S. uvarum*, a cross between strains isolated from North America (natural cross) and a cross between strains isolated from Europe (fermentation cross). We detected COs and NCOs across the 16 nuclear chromosomes of *S. uvarum*. Genomewide, we found significantly more COs on average in the natural cross (82.54 COs/meiosis, SE 1.5; 0.72 cM/kb) than in the fermentation cross (63.66 COs/meiosis, SE 1.9; 0.55 cM/kb; Wilcoxon rank sum test, p -value $\ll 0.001$) and significantly fewer NCOs in the natural cross (25.48 NCOs/meiosis, SE 0.99) than the fermentation cross (34.85 NCOs/meiosis, SE 8.92; Wilcoxon rank sum test, p -value=0.001448). The number of COs per meiosis in the natural and fermentation crosses are slightly higher than those of particular strains of *S. paradoxus* (54.8) and *S. cerevisiae* (76.5) respectively (Liu et al., 2019). Additionally, the average number of NCOs in the natural cross is lower than those of *S. cerevisiae* (46.4) and *S. paradoxus* (26.9), while the average number of NCOs in the fermentation cross is in between these values (Figure 2A; Tables S4 & S5; Liu et al., 2019). It's important to note that there is known strain variability in CO and NCO counts per meiosis for *S. cerevisiae* (CO: 90.5, 76.3, 73; NCO: 46.6, 46.4, 27; Mancera et al., 2008; Liu et al., 2019; Martini et al., 2011), similar to differences between strains of *S. uvarum* reported here.

To further explore the differences in recombination landscapes between our crosses, we split the genome into 20kb, non-overlapping windows, and obtained CO, NCO, and marker counts for each region. We downsampled markers in the introgression of the fermentation cross and found a significant effect of marker density on NCO count and tract length for most of the regions (Figures S2-10, Table S10). Because of how correlated marker density, NCO count, and introgression are, disentangling their relationships is complicated. For this reason, we applied an especially stringent correction for marker density—we used a previously published simulation based method for correction (Liu et al., 2019; Wijnker et al., 2013) and took the average NCO tract length (550bp) from introgression, where our marker density is highest, as the correction factor. For each window across the genome, we established our expected probability of detecting an NCO of that length and divided our observed NCO count by our probability of detecting an NCO event (see Methods). After the correction we found modest—but significant—genomewide correlation between our crosses for both COs (Spearman's correlation: 0.27; $p < 0.0001$) and NCOs (Spearman's correlation: 0.13; $p = 0.0018$).

We hypothesized that the low correlations between crosses might be impacted by the presence of heterozygous introgression from *S. eubayanus* in the fermentation cross. To explore this possibility, we separated the 20kb windows into introgressed and non-introgressed windows (based on whether they overlapped with an introgressed region). We will refer to introgression in the fermentation cross as “introgression” and use the term “introgressed region” to refer generally to the syntenic region, regardless of which cross we are focusing on. We calculated Spearman's correlations of COs, NCOs, and marker counts between the two crosses for each chromosome. We find positive correlation coefficients between CO counts between the fermentation and natural crosses for all chromosomes when looking at non-introgressed regions—though only chromosome 11 was significant (Table S6). For introgressed regions, we found no significant CO correlations between crosses and no consistency in the direction of the correlations, which is consistent with the hypothesis that CO landscapes are changed in introgression (Table S7). There were no significant correlations among NCOs. This was likely affected by the fact that markers used to detect NCOs are differently distributed between the crosses, and the small size of NCO tracts means that the marker resolution of a given region will affect the reported location of NCOs. This is because NCOs are reported to be at the midpoint of the markers that identify it; more dense markers will give a more precise location, which may conflict with the location reported by less dense markers. We find that introgressions tend to have lower CO counts and higher NCO counts in the fermentation cross when compared to syntenic regions in the natural cross (Wilcoxon rank

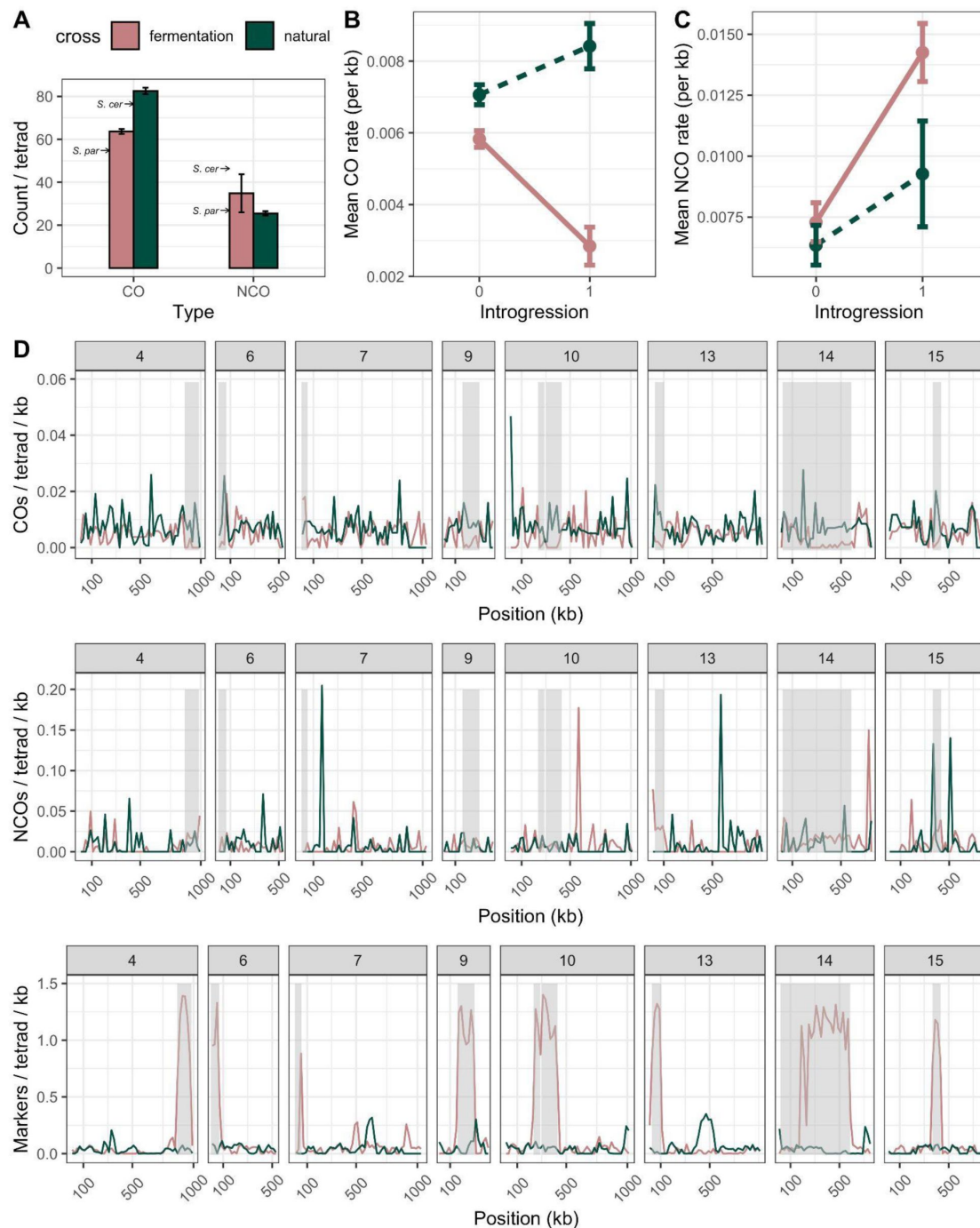


Figure 2

(A) Barplot depicting the number of COs and NCOs detected per meiosis in *S. uvarum* crosses (green: natural cross; pink: fermentation cross). The error bars represent the standard error around the mean. NCO counts are corrected for marker resolution. The counts for *S. paradoxus* and *S. cerevisiae* are represented by arrows and taken from Liu et al. (2019). (B) Mean CO/kb and (C) NCO/kb by cross and introgression (0 denotes intervals without introgression; 1 denotes introgression present in the fermentation cross. While the natural cross does not contain introgression, the region where introgression is present in the fermentation cross was compared to its syntenic region in the natural cross). NCO counts are corrected for marker resolution. Error bars represent the standard error around the mean. (D) *S. uvarum* chromosomes containing introgressions split into 20kb, non-overlapping windows. CO, NCO, and SNP counts are reported for both crosses (fermentation and natural). Shaded regions denote introgressed regions. CO counts are smoothed when the true location of the CO split could be in one of multiple windows. NCO counts are corrected for marker resolution.

sum test; CO p-value $<10^{-5}$, NCO p-value $<10^{-5}$). The fermentation cross has fewer COs than the natural cross overall, but the difference is greater in the introgressed regions (**Figure 2B-C**). Additionally, introgressed regions in the natural cross show significantly more COs and NCOs (Wilcoxon rank sum test; CO p-value=0.007, NCO p-value=0.006), indicating that these regions may be predisposed to having more recombination events even in the absence of introgression.

To further explore and test possible explanations for the patterns of COs and NCOs in introgressed regions, we constructed linear models for CO and NCO counts. Our model of CO counts showed a significant positive effect of the interactions between natural cross and introgression (whether a genomic window is from the natural cross and whether that window is in an introgressed region), as well as a significant negative effect of introgression on the number of COs, and a significant positive effect of GC content on CO count (**Table 1**). These results are consistent with increased COs in GC-rich regions and reduced COs in introgressions. Our model of NCO counts showed a similar positive effect of GC content on NCO counts, but showed an opposite significant coefficient for introgression and no significant effect of the interaction between introgression and cross—which was removed from the final model (**Table 2**). This indicates that GC content still plays an important role in localizing NCOs, and supports our findings that patterns of NCOs in introgressed regions are opposite to those of COs. Furthermore, the lack of a significant effect of cross echoes our finding of increased NCOs in introgressed regions in the natural cross.

Reduced diploid sequence similarity helps explain non-crossover repair of DSBs in introgression

One possible explanation for an increase of NCOs in introgression is that the reduced sequence similarity is biasing DSBs in the region to be repaired as NCOs rather than COs. We would therefore expect to see NCOs to be negatively correlated with sequence similarity. To evaluate the relationship of sequence similarity to the CO and NCO landscapes in introgressions, we measured diploid sequence similarity (the proportion of bases that are expected to match when we sample one base from each of the two parental strains), NCO depth, and CO count in 101 bp sliding windows with 50 bp overlaps along each of the introgressions. Mismatch repair proteins in *Saccharomyces* seem to suppress COs with very little mismatch in small regions (~350 bp), which informed our window size (Chen & Jinks-Robertson, 1999). We counted the number of NCO tracts that intersect with each window as a measurement of NCO depth, and simply counted the CO events in a given window (**Figure 3**). We then ran Spearman's correlations and a loess regression along each introgression and found a weak, but often significant ($p < 0.001$) correlation between NCOs and sequence similarity in the introgression (**Table 3**), suggesting that repair of double strand breaks is biased towards NCOs when sequence similarity is low. From the loess regression, we can observe an increase in NCO as sequence similarity reduces until about 0.9–0.8 sequence similarity, at which point NCOs level out or reduce (**Figure S1**). However, this effect is very weak with respect to the NCO counts, and at low levels of sequence similarity the uncertainty of the regression line is very large. This is primarily driven by the number of windows with no NCOs.

The low CO count in introgression leaves us unable to investigate effects of sequence similarity on CO counts except on chromosomes 7 and 14, where we found significantly higher sequence similarity around COs than around NCOs (**Table 4**). The introgressions on these two chromosomes are unique in that they contain a highly homologous portion of sequence and therefore contain enough COs for us to have power to detect differences between CO and NCO neighborhoods (**Figure 3**).

Table 1**Coefficients of gaussian generalized linear model modeling CO counts per 20kb window.**

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-8.9355	1.4713	-6.0732	1.70e-09
Introgression	-3.6343	0.6293	-5.7755	9.84e-09
GC	37.9733	3.7037	10.2528	< 2e-16
Introgression:crossnatural	5.2378	0.8595	6.0941	1.49e-09

Table 2**Coefficients of gaussian generalized linear model modeling NCO counts per 20kb window.**

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-11.4763	4.7810	-2.4004	0.0165
Introgression	4.3490	1.4937	2.9115	0.0037
GC	45.2894	12.0353	3.7631	0.0002

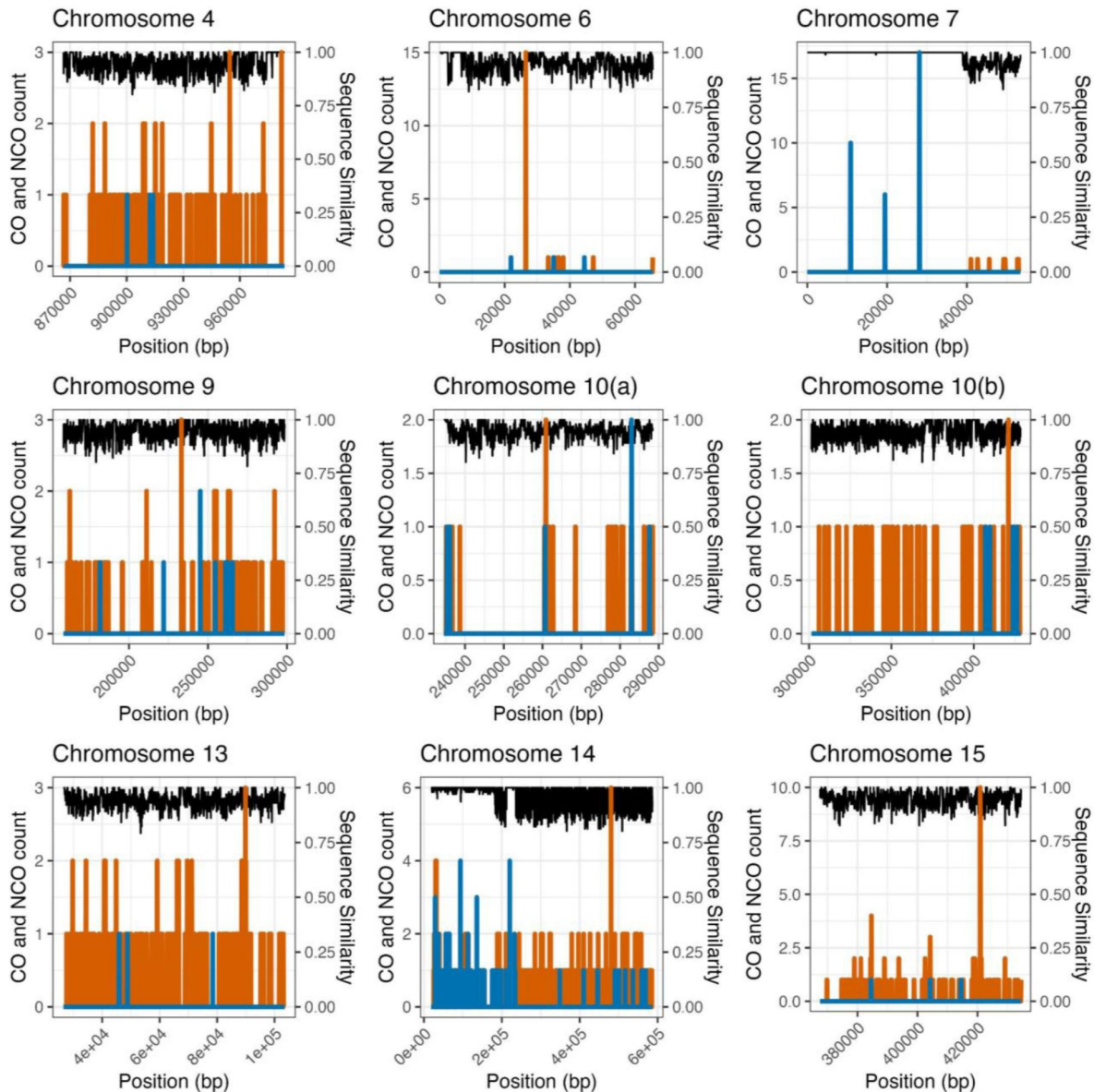


Figure 3

CO, NCO, and sequence similarity in 101bp sliding windows with 50bp overlaps of fermentation cross introgression. CO counts are shown in blue, the depth of NCO tracts are shown in orange, and the proportion of expected homologous bases between the two fermentation strains is shown in black.

Table 3

Spearman’s correlations of NCOs to sequence similarity in introgression in the fermentation cross. A cutoff p-value of 0.001 was selected by calculating the Bonferroni corrected $\alpha=0.05$ for nine comparisons (0.0011) and rounding down.

Chromosome	Start	End	Corr	p<0.001?
4	866500	983774	-0.1686	TRUE
6	1	65500	-0.0394	FALSE
7	1	53500	-0.2435	TRUE
9	158500	298500	-0.1460	TRUE
10	234500	288500	0.0073	FALSE
10	301500	428500	-0.1097	TRUE
13	26500	103500	-0.1982	TRUE
14	18500	586500	-0.2300	TRUE
15	367500	434500	-0.2328	TRUE

Introgression	CO mean		NCO mean		p-value
	sequence		sequence		
	similarity	CO SE	similarity	NCO SE	
Chromosome 4	0.9467	0.0148	0.9248	0.0029	0.2835
Chromosome 6	0.9663	0.0203	0.9784	0.0020	0.5594
Chromosome 7	1.0000	0.0000	0.9838	0.0018	<0.0001
Chromosome 9	0.9650	0.0102	0.9312	0.0037	0.0084
Chromosome 10 (1)	0.9671	0.0092	0.9476	0.0066	0.0419
Chromosome 10 (2)	0.9400	0.0074	0.9276	0.0040	0.1575
Chromosome 13	0.9700	0.0161	0.9304	0.0025	0.1292
Chromosome 14	0.9893	0.0026	0.9415	0.0018	<0.0001
Chromosome 15	0.9317	0.0093	0.9252	0.0031	0.5088

Table 4

Welch two sample t-test results for differences in sequence similarity between CO-adjacent regions and NCO-adjacent regions per introgression.

Introgression decreases allele shuffling locally and at the chromosome level

Because NCOs still play a small role in shuffling alleles along the chromosome, we were interested in whether the increase in NCOs of the fermentation cross would supplement the lost shuffling from the suppression of COs in the introgressions. To test this hypothesis, we used the measure r , which accounts for the number and positioning of recombination events to estimate the probability that a randomly chosen pair of loci shuffles their alleles in a gamete (Veller et al., 2019 [↗](#)). We calculated the average r per chromosome and for each introgressed region for each of the two crosses. We observed high levels of shuffling at the chromosome level when compared to humans. The intra-chromosomal component of r in humans is 0.0135 in females and 0.0177 in males (Veller et al., 2019 [↗](#)), while our measurements for chromosomes varied between 0.216 and 0.426. We find that most chromosomes do not have a significantly different amount of allele shuffling between the two crosses, even though the natural cross generally has more COs (Table S8; Bonferroni-adjusted $\alpha = 0.00313$). However, of the six chromosomes with significantly different r values, all of them showed more shuffling in the natural cross, and five of the six (chromosomes 4, 9, 10, 14, and 15) contained introgressed regions (Figure 4 [↗](#)). Lower r is not observed when introgressions are small and near telomeres, while even a small introgression near the center of the chromosome can lead to a large reduction in r (as is the case for chromosome 15). Chromosome 12 was the only chromosome without an introgressed region to have significantly different shuffling between crosses, and it also showed more shuffling in the natural cross. While it is unclear what potential mechanism is mediating the difference in shuffling on chromosome 12, we note that the rDNA locus on chromosome 12 is known to differ dramatically in repeat content across strains of *S. cerevisiae* (22–227 copies) (Sharma et al., 2022 [↗](#)), and we speculate that differences in rDNA copy number between strains in our crosses could impact shuffling. All of the introgressed regions showed significantly more shuffling in the natural cross, indicating that the large increase of NCOs in the introgressions does not make up for the loss of shuffling from the depletion of COs (Table S9; Bonferroni-adjusted $\alpha = 0.00556$). This finding indicates that an introgression that is segregating in a population will incur a shuffling cost in heterozygous individuals on top of any other evolutionary effects the introgression may have.

Discussion

Our study is motivated by understanding recombination rate variation within a species and uncovering potential genetic factors underlying this variation. To investigate this question, we crossed two pairs of *S. uvarum* strains, one pair isolated from natural environments and one pair from fermentation environments, and explored the distribution of CO and NCO events from both crosses. We detected more COs and fewer NCOs in our natural cross when compared to our fermentation cross, within a similar range of COs and NCOs per meiosis as previous studies in *S. cerevisiae* and *S. paradoxus* (Liu et al., 2019 [↗](#); Mancera et al., 2008 [↗](#)). Our findings demonstrate significant differences in recombination rate between closely related strains, adding to a body of literature across plants, animals, and fungi that recombination rate can evolve rapidly (Bauer et al., 2013 [↗](#); Danguy des Déserts et al., 2021 [↗](#); Kong et al., 2014 [↗](#); McGaugh et al., 2012 [↗](#); Petit et al., 2017 [↗](#); Raffoux et al., 2018 [↗](#); Samuk et al., 2019 [↗](#); Sandor et al., 2012 [↗](#); Schreiber et al., 2022 [↗](#)).

We hypothesized that these differences in the recombination landscape between our crosses were in part influenced by introgression, given that heterozygous introgression creates sequence divergence, and that COs in regions of heterozygosity are known to be curtailed in yeast and other organisms (Chen & Jinks-Robertson, 1999 [↗](#); Cooper et al., 2021 [↗](#); Martini et al., 2011 [↗](#)). We therefore explored the relationship between introgressions and the differences in CO and NCO

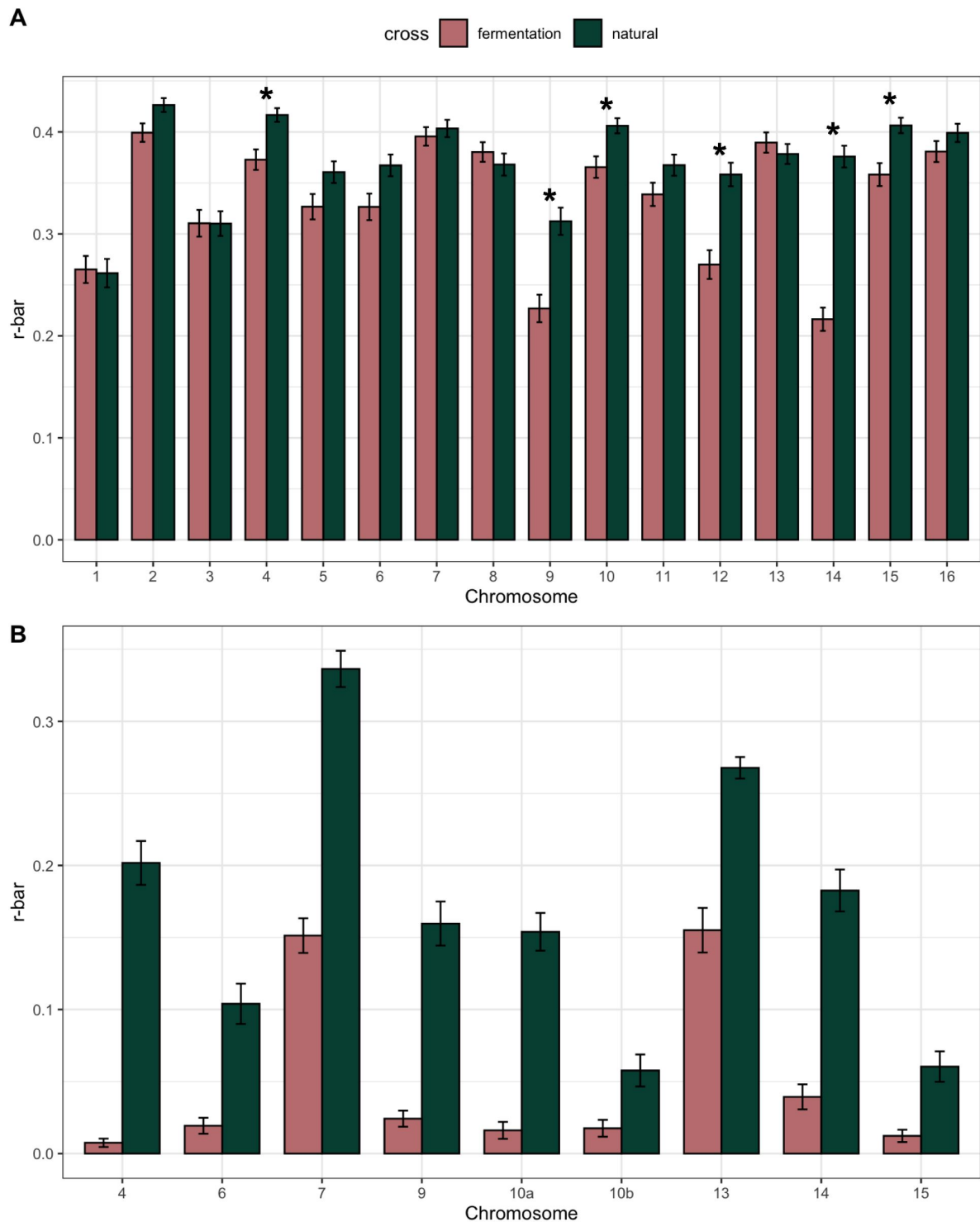


Figure 4

Average r for each chromosome (A) and for each introgressed region (B). Asterisks indicate a significant difference in chromosome r -bar between crosses. All introgressed regions had a significant difference in r . Error bars indicate standard error around the mean.

counts between crosses. We modeled CO and NCO locations, correcting for GC content (a well characterized driver of recombination events (Kiktev et al., 2018 [DOI](#); Marsolier-Kergoat & Yeramian, 2009)) and found that the distribution of COs and NCOs we observed could be partly explained by introgressions. While we are limited in our interpretations by only comparing two crosses (one cross with heterozygous introgression and one without introgression), these results are in line with findings in inversions, where heterozygotes show sharp decreases in COs, but the presence of NCOs in the inverted region (Crown et al., 2018 [DOI](#); Korunes & Noor, 2019 [DOI](#)). However, unlike heterozygous inversions where an increase in COs is observed on freely recombining chromosomes (the inter-chromosomal effect), we do not see an increase in COs on the borders flanking introgression or on chromosomes without introgression. Relatedly, our results also differ from observations in Arabidopsis, in which COs are elevated in regions with more heterozygosity at the cost of COs in adjacent homologous regions (Ziolkowski et al., 2015 [DOI](#); Ziolkowski & Henderson, 2017 [DOI](#)). These contrasting outcomes are likely in part due to the difference between intraspecific levels of polymorphism in Arabidopsis studies compared to the high level of inter-specific sequence divergence in our heterozygous introgression, although it also appears that mismatch repair gene MSH2 has functionally diverged between Arabidopsis and Saccharomyces (Blackwell et al., 2020 [DOI](#); Dlużewska et al., 2023 [DOI](#); Szymanska-Lejman et al., 2023). Regardless, it is curious that we do not find evidence for compensation of the depletion of COs due to introgression; the underlying mechanism responsible requires further study.

The introgressions present in the fermentation cross strains provided a unique opportunity to analyze NCO patterns, as the highly increased marker density lends the ability to detect NCOs with smaller tract lengths. We find evidence that DSB repair is biased towards NCOs when sequence similarity is low, with a weak but significant effect of increasing NCOs with decreasing sequencing similarity to a peak at sequence similarity between 0.8-0.9. Our data are thus aligned with previous work in *S. cerevisiae* and other systems, in which *MSH2* and *SGS1* act as anti-recombinases and disassemble D-loops with mismatched sequences, resulting in NCOs instead of COs (Borts & Haber, 1987 [DOI](#); Chen & Jinks-Robertson, 1999 [DOI](#); Martini et al., 2011 [DOI](#); Mazina et al., 2004 [DOI](#); Myung et al., 2001 [DOI](#)). By downsampling markers in the introgressed regions, we found an increase in the number of NCOs and a decrease in their tract length as we increased our resolution (Figures S2-S10). Our expectation was to find a marker density at which NCO counts and their tract lengths would no longer increase and decrease respectively. However, we didn't find such a density, despite in some cases exploring marker densities that included all the markers in the introgression. This leaves us with the uncertainty of whether NCOs in the introgressions are more frequent and smaller than in other regions, or if we may be underestimating NCO counts and overestimating their tract lengths in cases where marker resolution is lower.

One likely effect of the CO reduction we observed in introgressions is a reduction in allele shuffling at the regional and chromosomal level. While NCOs can increase local shuffling, they likely have a much weaker effect on the likelihood of two random alleles being shuffled than COs do. We find this is the case for our two crosses, where despite a large number of NCOs in introgressions, the amount of shuffling (as measured by r) is significantly lower in the fermentation cross. This loss of shuffling translates to frequently lower r in the fermentation cross at the chromosome level for chromosomes containing introgressions. The exceptions being small introgressions near the telomeres, which is consistent with the expectation that COs near the center of chromosomes generate much more shuffling of alleles than terminal COs (Veller et al., 2019 [DOI](#)). Our findings indicate that reducing COs, especially near the center of chromosomes, has a cost to shuffling that is not compensated by the increase of NCOs that we observe. If the benefit of recombination is its ability to generate new combinations of alleles, then the loss of shuffling resulting from being heterozygous for divergent DNA sequences may come at an additional cost beyond the possibilities of genetic incompatibilities between hybridizing species. This cost is likely higher as divergence increases and as the length of divergent sequences is greater, as is the case

with early generation hybrids (Dagilis & Matute, 2023 [↗](#)). Ultimately, if sequence divergence is too high, the resultant failure to recombine can become a postzygotic reproductive barrier (Bozdogan et al., 2021 [↗](#); Hunter et al., 1996 [↗](#); Rogers et al., 2018 [↗](#)).

The shuffling cost to introgression that we identify in our crosses may play an important role in the fate of introgression in the generations following hybridization. When heterozygotes for an introgression are formed, the reduction in shuffling inside the introgression will increase the likelihood that the introgression is purged from the population. This is because it will likely be inherited in its entirety and will carry the fitness cost of incompatibilities combined with a cost of shuffling. This cost is incurred because the reduction of COs in the introgression will reduce shuffling of alleles on either side of it and will vary in its intensity depending on the location and size of the introgression. In generations immediately following hybridization, introgressions will be much larger and are therefore expected to be more costly (although this likely depends on a number of factors including time since divergence). These predictions are consistent with modeling and empirical data on the purging of introgression in *Drosophila* and humans in the first generations following hybridization (Veller et al., 2023 [↗](#)).

As to longer term dynamics of recombination and selection, we predict that the excess NCOs detected in heterozygous introgressions should begin to erode the divergence between the sequences, increasing homology and slowly reducing the cost of the introgression. This hypothesis posits that recombination can act to remove the larger, more deleterious regions of an introgression quickly while whittling away slightly deleterious alleles that may be linked to any beneficial regions of an introgression. While our current study doesn't capture longer term patterns of recombination or the landscape of recombination in introgressions that would lead to introgressions preferentially remaining in high-CO regions, it's notable that introgressed regions coincide with high-recombination regions in the natural cross, indicating that these regions may be primed to retain introgressions after a hybridization event—possibly due to the increased shuffling allowing for a larger fraction of the introgression to remain (Veller et al., 2023 [↗](#)).

Finally, we note that *Saccharomyces* typically reproduce asexually, with only infrequent sexual cycles (Magwene et al., 2011 [↗](#); Ruderfer et al., 2006 [↗](#); Zeyl & Otto, 2007 [↗](#)). When they do mate, they often mate within a tetrad resulting in increased homozygosity. For example, each diploid progenitor of the parents of our fermentation cross was homozygous for introgression across the genome, meaning that recombination would neither break up nor aid in purging the introgression in isolated populations of each parent. This suggests that the fate of introgressions in this species is perhaps more loosely tied to recombination patterns than it would be in an obligately sexually reproducing species.

Despite some limitations to interpretation, this study provides a unique view of the early dynamics of hybridization and the role of recombination in the presence of introgression. By focusing not only on the distribution of recombination events but on their specific role in shuffling alleles, we can more closely connect the physical process of recombination to its role among other evolutionary forces.

Methods

Strain and library construction

S. uvarum strains (UCD61-137, yHCT78, GM14, and DBVPG7787) were obtained from the Portuguese Yeast Culture Collection and from Chris Hittinger (Table S1) (Almeida et al., 2014 [↗](#)). All four *S. uvarum* strains had their *HO* locus replaced with a kanMX marker using a modified version of the high-efficiency yeast transformation using the LiAc/SS carrier DNA/PEG method. Briefly, the kanMX marker was amplified from plasmid pCSH2 with homology to genomic DNA flanking the

HO ORF with primers CSH239

(GGTGGAAAACACGAAAAGTTAGAACTACGTTTCAGGCAAAgacatggaggccagaatac) and CSH241
(GTGACCGTATTGGTACTTTTTTTGTTACCTGTTTTAGTAGcagtatagcgaccagcattc).

For each strain, overnight cultures were inoculated in 25 mL of YPD at an OD of ~ 0.0005 and incubated at room temperature on a shaker for ~24 hours until the cultures reached an OD between 0.6 and 0.9. Subsequently, 1 ug of the template DNA was transformed with a heat shock temperature of 37°C for 45 minutes. The transformed cells were allowed to recover in liquid YPD for 4 hours before being plated onto G418 selective plates and incubated at room temperature for 2 days.

Single colonies were selected from the transformation plates, restreaked onto G418 plates and allowed to grow at room temperature for 2 days. Single colonies from those plates were then inoculated into 2 mL of YPD + G418 and incubated in a roller drum at room temperature overnight. From those cultures, 250 uL was used to inoculate 2 mL of sporulation media (1% potassium acetate, 0.1 % yeast extract, 0.05% dextrose) and incubated at room temperature for 3 to 5 days. Strains were confirmed to have the *ho::KanMX* via tetrad dissection on a Singer SporPlay+ microscope (Singer Instruments). Plates with tetrads were incubated at room temperature for 2 days and then replica plated to test for proper segregation of the *kanMX* marker and mating type within individual tetrads.

Crosses between strains UCD61-137 and yHCT78 (natural cross), and between strains GM14 and DBVPG7787 (fermentation cross) were set up by micromanipulation of single MATa and MATx cells using a Singer SporPlay+. The plates were incubated at room temperature for 2 days and then replica plated to mating type tester strains to test for potential diploids. Identified diploids were then sporulated by growing a culture of the cross in 2 mL YPD + G418 at room temperature overnight. From those cultures, 250 uL were used to inoculate 2 mL of sporulation media and incubated at room temperature for 3 to 5 days. Sporulated cultures were dissected on 3 YPD plates (24 tetrads per plate) using a Singer SporPlay+. Fifty of the fully viable tetrads were selected and had all their spores inoculated into YPD (200 spores total) and incubated at room temperature. The DNA was extracted from these cultures using a modified version of the Hoffman-Winston DNA Prep (Hoffman & Winston, 1987 [\[1\]](#)). The DNA concentration was then measured using SYBR green, and 150 ng of each sample's DNA was used to prepare a sequencing library using an Illumina DNA Prep Kit, modified to use half the normal amounts of reagents. Libraries were pooled and run on an Illumina NovaSeq 500 with 150bp paired end reads.

Calling SNPs

We scored SNPs from parents and offspring using the *S. uvarum* reference genome (Scannell et al., 2011 [\[2\]](#)) and custom scripts that invoked bwa (v0.7.17), samtools (v1.12), bcftools (v1.13), picard tools (v2.25.6), and gatk (v4.2.0.0) (Danecek et al., 2021 [\[3\]](#); Li & Durbin, 2009 [\[4\]](#); McKenna et al., 2010 [\[5\]](#)). The custom scripts are available in the github repository: *schwarzkopf/CO-NCO*. We joint genotyped parents and offspring with default filters for gatk with the exception of the QUAL filter, which was set as < 100 for parents and < 30 for offspring. We further filtered variants by requiring they be fixed differences between the two parental strains. We kept a total of 24,574 markers for the natural cross and 74,619 markers for the fermentation cross. We utilized LUMPY to identify structural variants in the parent strains that were greater than 5000 bp and verified calls using the Integrative Genomics Viewer (Layer et al., 2014 [\[6\]](#); Robinson et al., 2011 [\[7\]](#)). We identified three amplifications in strain GM14 (one of the fermentation cross parents) that were absent in other strains (Table S2).

Generating CO/NCO maps

We generated “seg” files by coding tetrad variants by their parental origin. These seg files were the input for CrossOver (v6.3) from the ReCombine suite of programs, which we used to detect COs and NCOs (Anderson et al., 2011 [↗](#)). We then filtered to remove non-crossovers with fewer than three associated markers and split the genome into 20kb windows. In each window we counted crossovers, non-crossovers, and markers. We established regions of introgressions through visual inspection of marker density in the fermentation cross (introgressions showed more divergence between fermentation strains) and confirmed them using the findings of Almeida et al. (2014) [↗](#). We found nine heterozygous introgressions on chromosomes 4, 6, 7, 9, 10, 10, 13, 14, and 15 respectively that we included in further analyses (Table S3). We excluded two additional introgressions due to poor mapping (chromosome 13:0-17,000; chromosome 16: 642,000-648,000). We analyzed the effect of marker resolution on NCO detection by randomly removing markers from each introgression and running CrossOver and its downstream counting process as described above. For each introgression, we downsampled to multiple percentiles of the distribution of non-introgressed region marker counts per kb (10th: 0.05; 25th: 0.75; 50th: 1.95; 75th: 2.95; 90th: 4.45; 95th: 7.87; 99th: 14.01) and the median marker count per kb for introgressed (3.225) and non-introgressed (1.35) regions. We repeated each downsampling amount 20 times for each introgression and extracted average CO counts, NCO counts, and NCO tract lengths. To account for the difference in number of markers in introgressed vs non-introgressed windows and their effect on NCO detection, we applied a previously published simulation-based method (Liu et al., 2019 [↗](#); Wijnker et al., 2013 [↗](#)). We chose the average NCO tract length of the fermentation cross's introgressed regions—550 bp—and for each window randomly placed an NCO event of that length 10,000 times to establish our expected probability of detecting an NCO of that length. We then divided our observed NCO count by our probability of detecting an NCO event. Additionally, because COs that occurred in large regions devoid of markers would be called in the middle of the empty windows, we split CO counts in regions with multiple consecutive marker-free windows evenly between the empty windows. With these corrected maps, we calculated spearman correlations between crosses using R (v4.1.0, R Core Team 2021). Additionally, we modeled NCO and CO count as a function of introgression, introgression by cross, and GC content using a gaussian generalized linear model in R (v4.1.0, R Core Team 2021).

Sequence similarity

We calculated diploid sequence similarity between the two fermentation cross strains in 101bp windows with 50bp overlaps. At each nucleotide position in the window, we counted fixed differences as zero diploid sequence similarity, invariant sites between strains as full diploid sequence similarity (1), and polymorphic sites in either or both strains as half diploid sequence similarity (0.5). We then averaged these sequence similarity values across the window. This measure represents the probability that both strains will have the same nucleotide base at a given position. We used this measure of fine-scale sequence similarity to determine how sequence similarity related to NCO counts in introgressed regions. For this, we used Loess regressions and Spearman's correlations on each of the introgressed regions comparing sequence similarity to NCO count, both implemented in R (v4.1.0, R Core Team 2021). We then focused on each recombination event (CO or NCO) and compared the sequence similarity 100bp up and downstream of CO breakpoints and 100bp up and downstream of NCO tracts.

We then used Welch's two sample t-tests to compare CO and NCO sequence similarity in each introgression.

r

We use r , a measure genetic shuffling defined in [Veller et al. \(2019\)](#) to measure how much shuffling occurs in each chromosome for each cross. Our data provides parental origin for each fixed difference between parental strains. We assume that all loci between pairs of markers that come from the same parent are also from that parent. We also assume that when a pair of successive markers come from different parents, the location of the change from one parental origin to the other happens at the midpoint between our markers. With this in mind, we counted the number of bases that come from one parent and divided by the chromosome size to obtain the proportion of the chromosome that was inherited from said parent (p) and used the formula from [Veller et al. \(2019\)](#): $r = 2p(1 - p)$. We calculated r for each full chromosome and each introgressed regions in every gamete from both crosses. We then averaged across gametes to obtain average r values. We then compared average r between crosses in each chromosome or introgressed region using Welch two sample t-tests and correcting for multiple tests using a Bonferroni correction in R (v4.1.0, R Core Team 2021).

Data Access

Sequences for the parental strains can be found on NCBI SRA (SRR1119189, SRR1119180 SRR1119199, SRR1119200) ([Almeida et al., 2014](#)). Sequencing of the tetrads is deposited at NCBI SRA under Project PRJNA1061120. Scripts are available in the github repository: *ejschwarzkopf/CO-NCO*.

Acknowledgements

We are grateful to members of the Heil lab, Mohamed Noor, Nathan Layman, Mark Smithson, and three anonymous reviewers for comments on this manuscript. We thank Chris Hittinger and the Portuguese Yeast Culture Collection for *S. uvarum* strains. This work was supported by NIH R35GM142849 to CSH.

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Editors

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Reviewer #1 (Public review):

Summary:

The authors explored how the presence of interspecific introgressions in the genome affects the recombination landscape. This research aims to shed light on the genetic phenomena influencing the evolution of introgressed regions. However, it is important to note that the study is based on examining only one generation, which limits the scope for making broad evolutionary conclusions. In this study, yeast hybrids with large introgressions (ranging from several to several dozen percent of the chromosome length) from another yeast species were crossed. The products of meiosis were then isolated and sequenced to examine the genome-wide distribution of both crossovers (COs) and noncrossovers (NCOs). The authors found a significant reduction in the frequency of COs within the introgressed regions, which is a phenomenon well-documented in various systems. They also report that introgressed regions exhibit an increased frequency of NCOs. Unfortunately, this conclusion seems flawed, as there is no accurate method for correcting the detection level of NCOs when the compared regions (introgressed and non-introgressed) differ drastically in SNP density. The authors further confirmed that introgressions significantly limit the local shuffling of genetic information, and while NCOs contribute slightly to this shuffling, they do not compensate for the loss of CO recombination. This is widely known fact.

In summary, the study makes a limited contribution to the understanding of how polymorphism impacts meiotic recombination. The conclusion regarding the increase in NCO frequency in polymorphic regions is likely incorrect.

<https://doi.org/10.7554/eLife.96184.2.sa2>

Reviewer #3 (Public review):

When members of two related but diverged species mate, the resulting hybrids can produce offspring where parts of one species' genome replace those of the other. These "introgressions" often create regions with a much greater density of sequence differences than are normally found between members of the same species. Previous studies have shown that increased sequence differences, when heterozygous, can reduce recombination during meiosis specifically in the region of increased difference. However, most of these studies have focused on crossover recombination, and have not measured noncrossovers. The current study uses a pair of *Saccharomyces uvarum* crosses: one between two natural isolates that, while exhibiting some divergence, do not contain introgressions; the other is between two

fermentation strains that, when combined, are heterozygous for 9 large regions of introgression that have much greater divergence than the rest of the genome. The authors wished to determine if introgressions differently affected crossovers and noncrossovers, and, if so, what impact that would have on the gene shuffling that occurs during meiosis.

While both crossovers and noncrossovers were measured, assessing the true impact of increased heterology (inherent in heterozygous introgressions) is complicated by the fact that the increased marker density in heterozygous introgressions also increases the ability to detect noncrossovers. The authors now use a revised correction aimed at compensating for this difference, and based on that correction, conclude that, while as expected crossovers are decreased by increased sequence heterology, noncrossovers neither increase nor decrease substantially. They then show that genetic shuffling overall is substantially reduced in regions of heterozygous introgression, which is not surprising given that one type of event is reduced and the other remains at similar levels. However, the correction currently used remains poorly justified, tests of its validity are not presented. Thus, the only possibly novel conclusion, that noncrossovers are less affected by heterology than crossovers, remains to be adequately tested.

In conclusion, of the three main conclusions as stated in the abstract, one (that crossovers go down) has been shown in many systems, one (that noncrossovers increase) is wrong, and the third (that allele shuffling is reduced) is obvious. Given this, the impact of this work on the field will be minimal at best, and negative to the extent that readers are led astray.

<https://doi.org/10.7554/eLife.96184.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The authors investigated how the presence of interspecific introgressions in the genome affects the recombination landscape. This research was intended to inform about genetic phenomena influencing the evolution of introgressed regions, although it should be noted that the research itself is based on examining only one generation, which limits the possibility of drawing far-reaching evolutionary conclusions. In this work, yeast hybrids with large (from several to several dozen percent of the chromosome length) introgressions from another yeast species were crossed. Then, the products of meiosis were isolated and sequenced, and on this basis, the genome-wide distribution of both crossovers (COs) and noncrossovers (NCOs) was examined. Carrying out the analysis at different levels of resolution, it was found that in the regions of introduction, there is a very significant reduction in the frequency of COs and a simultaneous increase in the frequency of NCOs. Moreover, it was confirmed that introgressions significantly limit the local shuffling of genetic information, and NCOs are only able to slightly contribute to the shuffling, thus they do not compensate for the loss of CO recombination.

Strengths:

- Previously, experiments examining the impact of SNP polymorphism on meiotic recombination were conducted either on the scale of single hotspots or the entire hybrid

genome, but the impact of large introgressed regions from another species was not examined. Therefore, the strength of this work is its interesting research setup, which allows for providing data from a different perspective.

- Good quality genome-wide data on the distribution of CO and NCO were obtained, which could be related to local changes in the level of polymorphism.

Weaknesses:

(1) The research is based on examining only one generation, which limits the possibility of drawing far-reaching evolutionary conclusions. Moreover, meiosis is stimulated in hybrids in which introgressions occur in a heterozygous state, which is a very unlikely situation in nature. Therefore, I see the main value of the work in providing information on the CO/NCO decision in regions with high sequence diversification, but not in the context of evolution.

While we are indeed only examining recombination in a single generation, we respectfully disagree that our results aren't relevant to evolutionary processes. The broad goals of our study are to compare recombination landscapes between closely related strains, and we highlight dramatic differences between recombination landscapes. These results add to a body of literature that seeks to understand the existence of variation in traits like recombination rate, and how recombination rate can evolve between populations and species. We show here that the presence of introgression can contribute to changes in recombination rate measured in different individuals or populations, which has not been previously appreciated. We furthermore show that introgression can reduce shuffling between alleles on a chromosome, which is recognized as one of the most important determinants for the existence and persistence of sexual reproduction across all organisms. As we describe in our introduction and conclusion, we see our experimental exploration of the impacts of introgression on the recombination landscape as complementary to studies inferring recombination and introgression from population sequencing data and simulations. There are benefits and challenges to each approach, but both can help us better understand these processes. In regards to the utility of exploring heterozygous introgression, we point out that introgression is often found in a heterozygous state (including in modern humans with Neanderthal and/or Denisovan ancestry). Introgression will always be heterozygous immediately after hybridization, and depending on the frequency of gene flow into the population, the level of inbreeding, selection against introgression, etc., introgression will typically be found as heterozygous.

- The work requires greater care in preparing informative figures and, more importantly, re-analysis of some of the data (see comments below).

More specific comments:

(1) The authors themselves admit that the detection of NCO, due to the short size of conversion tracts, depends on the density of SNPs in a given region. Consequently, more NCOs will be detected in introgressed regions with a high density of polymorphisms compared to the rest of the genome. To investigate what impact this has on the analysis, the authors should demonstrate that the efficiency of detecting NCOs in introgressed regions is not significantly higher than the efficiency of detecting NCOs in the rest of the genome. If it turns out that this impact is significant, analyses should be presented proving that it does not entirely explain the increase in the frequency of NCOs in introgressed regions.

We conducted a deeper exploration of the effect of marker resolution on NCO detection by randomly removing different proportions of markers from introgressed regions of the fermentation cross in order to simulate different marker resolutions from non-introgressed

regions. We chose proportions of markers that would simulate different quantiles of the resolution of non-introgressed regions and repeated our standard pipeline in order to compare our NCO detection at the chosen marker densities. More details of this analysis have been added to the manuscript (lines 188-199, 525-538). We confirmed the effect of marker resolution on NCO detection (as reported in the updated manuscript and new supplementary figures S2-S10, new Table S10) and decided to repeat our analyses on the original data with a more stringent correction. For this we chose our observed average tract size for NCOs in introgressed regions (550bp), which leads to a far more conservative estimate of NCO counts (As seen in the updated Figure 2 and Table 2). This better accounts for the increased resolution in introgressed regions, and while it's possible to be more stringent with our corrections, we believe that further stringency would be unreasonable. We also see promising signs that the correction is sufficient when counting our CO and NCO events in both crosses, as described in our response to comment 39 (response to reviewer #3).

(2) CO and NCO analyses performed separately for individual regions rarely show statistical significance (Figures 3 and 4). I think that the authors, after dividing the introgressed regions into non-overlapping windows of 100 bp (I suggest also trying 200 bp, 500 bp, and 1kb windows), should combine the data for all regions and perform correlations to SNP density in each window for the whole set of data. Such an analysis has a greater chance of demonstrating statistically significant relationships. This could replace the analysis presented in Figure 3 (which can be moved to Supplement). Moreover, the analysis should also take into account indels.

We're uncertain of what is being requested here. If the comment refers to the effect of marker density on NCO detection, we hope the response to comment 2 will help resolve this comment as well. Otherwise, we ask for some clarification so that we may correct or revise as appropriate.

(3) In Arabidopsis, it has been shown that crossover is stimulated in heterozygous regions that are adjacent to homozygous regions on the same chromosome (<http://dx.doi.org/10.7554/eLife.03708.001>, <https://doi.org/10.1038/s41467-022-35722-3>).

This effect applies only to class I crossovers, and is reversed for class II crossovers (<https://doi.org/10.15252/embj.2020104858>, <https://doi.org/10.1038/s41467-023-42511-z>). This research system is very similar to the system used by the authors, although it likely differs in the level of DNA sequence divergence. The authors could discuss their work in this context.

We thank the reviewer for sharing these references. We have added a discussion of our work in the context of these findings in the Discussion, lines 367-376.

Reviewer #2 (Public Review):

Summary:

*Schwartzkopf et al characterized the meiotic recombination impact of highly heterozygous introgressed regions within the budding yeast *Saccharomyces uvarum*, a close relative of the canonical model *Saccharomyces cerevisiae*. To do so, they took advantage of the naturally occurring *Saccharomyces bayanus* introgressions specifically within fermentation isolates of *S. uvarum* and compared their behavior to the syntenic regions of a cross between natural isolates that do not contain such introgressions. Analysis of crossover (CO) and noncrossover (NCO) recombination events shows both a depletion in CO frequency within highly heterozygous introgressed regions and an increase in NCO frequency. These results strongly support the hypothesis that DNA sequence polymorphism inhibits CO formation, and has no or much weaker effects on*

NCO formation. Eventually, the authors show that the presence of introgressions negatively impacts " r ", the parameter that reflects the probability that a randomly chosen pair of loci shuffles their alleles in a gamete.

The authors chose a sound experimental setup that allowed them to directly compare recombination properties of orthologous syntenic regions in an otherwise intra-specific genetic background. The way the analyses have been performed looks right, although this reviewer is unable to judge the relevance of the statistical tests used. Eventually, most of their results which are elegant and of interest to the community are present in Figure 2.

Strengths:

Analysis of crossover (CO) and noncrossover (NCO) recombination events is compelling in showing both a depletion in CO frequency within highly heterozygous introgressed regions and an increase in NCO frequency.

Weaknesses:

The main weaknesses refer to a few text issues and a lack of discussion about the mechanistic implications of the present findings.

- Introduction

(1) The introduction is rather long. | I suggest specifically referring to "meiotic" recombination (line 71) and to "meiotic" DSBs (line 73) since recombination can occur outside of meiosis (ie somatic cells).

We agree and have condensed the introduction to be more focused. We also made the suggested edits to include "meiotic" when referring to recombination and DSBs.

(2) From lines 79 to 87: the description of recombination is unnecessarily complex and confusing. I suggest the authors simply remind that DSB repair through homologous recombination is inherently associated with a gene conversion tract (primarily as a result of the repair of heteroduplex DNA by the mismatch repair (MMR) machinery) that can be associated or not to a crossover. The former recombination product is a crossover (CO), the latter product is a noncrossover (NCO) or gene conversion. Limited markers may prevent the detection of gene conversions, which erase NCO but do not affect CO detection.

We changed the language in this section to reflect the reviewer's suggestions.

(3) In addition, "resolution" in the recombination field refers to the processing of a double Holliday junction containing intermediates by structure-specific nucleases. To avoid any confusion, I suggest avoiding using "resolution" and simply sticking with "DSB repair" all along the text.

We made the suggested correction throughout the paper.

(4) Note that there are several studies about *S. cerevisiae* meiotic recombination landscapes using different hybrids that show different CO counts. In the introduction, the authors refer to Mancera et al 2008, a reference paper in the field. In this paper, the hybrid used showed ca. 90 CO per meiosis, while their reference to Liu et al 2018 in Figure 2 shows less than 80 COs per meiosis for *S. cerevisiae*. This shows that it is not easy to come up with a definitive CO count per meiosis in a given species. This needs to be taken into account for the result section line 315-321.

This is an excellent point. We added this context in the results (lines 180-187).

(5) In line 104, the authors refer to S. paradoxus and mention that its recombination rate is significantly different from that of S. cerevisiae. This is inaccurate since this paper claims that the CO landscape is even more conserved than the DSB landscape between these two species, and they even identify a strong role played by the subtelomeric regions. So, the discussion about this paper cannot stand as it is.

We agree with the reviewer's point. We also found that the entire paragraph was unnecessary, so it and the sentence in question have been removed.

(6) Line 150, when the authors refer to the anti-recombinogenic activity of the MMR, I suggest referring to the published work from Martini et al 2011 rather than the not-yet-published work from Copper et al 2021, or both, if needed.

Added the suggested citation.

Results

(7) The clear depletion in CO and the concomitant increase in NCO within the introgressed regions strongly suggest that DNA sequence polymorphism triggers CO inhibition but does not affect NCO or to a much lower extent. Because most CO likely arises from the ZMM pathway (CO interference pathway mainly relying on Zip1, 2, 3, 4, Spo16, Msh4, 5, and Mer3) in S. uvarum as in S. cerevisiae, and because the effect of sequence polymorphism is likely mediated by the MMR machinery, this would imply that MMR specifically inhibits the ZMM pathway at some point in S. uvarum. The weak effect or potential absence of the effect of sequence polymorphism on NCO formation suggests that heteroduplex DNA tracts, at least the way they form during NCO formation, escape the anti-recombinogenic effect of MMR in S. uvarum. A few comments about this could be added.

We have added discussion and citations regarding the biased repair of DSB to NCO in introgression, lines 380-386.

(8) The same applies to the fact that the CO number is lower in the natural cross compared to the fermentation cross, while the NCO number is the same. This suggests that under similar initiating Spo11-DSB numbers in both crosses, the decrease in CO is likely compensated by a similar increase in inter-sister recombination.

Thank you to the reviewer for this observation. We agree that this could explain some differences between the crosses.

(9) Introgressions represent only 10% of the genome, while the decrease in CO is at least 20%. This is a bit surprising especially in light of CO regulation mechanisms such as CO homeostasis that tends to keep CO constant. Could the authors comment on that?

We interpret these results to reflect two underlying mechanisms. First, the presence of heterozygous introgression does reduce the number of COs. Second, we believe the difference in COs reflects variation in recombination rate between strains. We note that CO homeostasis need not apply across different genetic backgrounds. Indeed, recombination rate is appreciated to significantly differ between strains of *S. cerevisiae* (Raffoux et al. 2018), and recombination rate variation has been observed between strains/lines/populations in many different species including *Drosophila*, mice, humans, *Arabidopsis*, maize, etc. We reference

S. cerevisiae strain variability in the Introduction lines 128-130, and have added context in the Results lines 180-187, and Discussion lines 343-350.

(10) Finally, the frequency of NCOs in introgressed regions is about twice the frequency of CO in non-introgressed regions. Both CO and NCO result from Spo11-initiating DSBs.

This suggests that more Spo11-DSBs are formed within introgressed regions and that such DSBs specifically give rise to NCO. Could this be related to the lack of homolog engagement which in turn shuts down Spo11-DSB formation as observed in ZMM mutants by the Keeney lab? Could this simply result from better detection of NCO in introgressed regions related to the increased marker density, although the authors claim that NCO counts are corrected for marker resolution?

The effect noted by the reviewer remains despite the more conservative correction for marker density applied to NCO counts (as described in the response to Reviewer 1, comment #2). Given that CO+NCO counts in introgressed regions are not statistically different between crosses, it is likely that these regions are simply predisposed to a higher rate of DSBs than the rest of the genome. This is an interesting observation, however, and one that we would like to further explore in future work.

(11) What could be the explanation for chromosome 12 to have more shuffling in the natural cross compared to the fermentation cross which is deprived of the introgressed region?

We added this text to the Results, lines 323-327, "While it is unclear what potential mechanism is mediating the difference in shuffling on chromosome 12, we note that the rDNA locus on chromosome 12 is known to differ dramatically in repeat content across strains of *S. cerevisiae* (22–227 copies) (Sharma et al. 2022), and we speculate that differences in rDNA copy number between strains in our crosses could impact shuffling."

Technical points:

(12) In line 248, the authors removed NCO with fewer than three associated markers.

What is the rationale for this? Is the genotyping strategy not reliable enough to consider events with only one or two markers? NCO events can be rather small and even escape detection due to low local marker density.

We trust the genotyping strategy we used, but chose to be conservative in our detection of NCOs to account for potential sequencing biases.

(13) Line 270: The way homology is calculated looks odd to this reviewer, especially the meaning of 0.5 homology. A site is either identical (1 homology) or not (0 homology).

We've changed the language to better reflect what we are calculating (diploid sequence similarity; see comment #28). Essentially, the metric is a probability that two randomly selected chromatids—one from each parent—will share the same nucleotide at a given locus (akin to calculating the probability of homozygous offspring at a single locus). We average it along a segment of the genome to establish an expected sequence similarity if/when recombination occurs in that segment.

(14) Line 365: beware that the estimates are for mitotic mismatch repair (MMR). Meiotic MMR may work differently.

We removed the citation that refers exclusively to mitotic recombination. The statement regarding meiotic recombination is otherwise still reflective of results from Chen & Jinks-Robertson

(15) Figure 1: there is no mention of potential 4:0 segregations. Did the authors find no such pattern? If not, how did they consider them?

The program we used to call COs and NCOs (ReCombine's CrossOver program) can detect such patterns, but none were detected in our data.

Reviewer #3 (Public Review):

*When members of two related but diverged species mate, the resulting hybrids can produce offspring where parts of one species' genome replace those of the other. These "introgressions" often create regions with a much greater density of sequence differences than are normally found between members of the same species. Previous studies have shown that increased sequence differences, when heterozygous, can reduce recombination during meiosis specifically in the region of increased difference. However, most of these studies have focused on crossover recombination, and have not measured noncrossovers. The current study uses a pair of *Saccharomyces uvarum* crosses: one between two natural isolates that, while exhibiting some divergence, do not contain introgressions; the other is between two fermentation strains that, when combined, are heterozygous for 9 large regions of introgression that have much greater divergence than the rest of the genome. The authors wished to determine if introgressions differently affected crossovers and noncrossovers, and, if so, what impact that would have on the gene shuffling that occurs during meiosis.*

(1) While both crossovers and noncrossovers were measured, assessing the true impact of increased heterology (inherent in heterozygous introgressions) is complicated by the fact that the increased marker density in heterozygous introgressions also increases the ability to detect noncrossovers. The authors used a relatively simple correction aimed at compensating for this difference, and based on that correction, conclude that, while as expected crossovers are decreased by increased sequence heterology, counter to expectations noncrossovers are substantially increased. They then show that, despite this, genetic shuffling overall is substantially reduced in regions of heterozygous introgression. However, it is likely that the correction used to compensate for the effect of increased sequence density is defective, and has not fully compensated for the ascertainment bias due to greater marker density. The simplest indication of this potential artifact is that, when crossover frequencies and "corrected" noncrossover frequencies are taken together, regions of introgression often appear to have greater levels of total recombination than flanking regions with much lower levels of heterology. This concern seriously undercuts virtually all of the novel conclusions of the study. Until this methodological concern is addressed, the work will not be a useful contribution to the field.

We appreciate this concern. Please see response to comments #2 and #38. We further note that our results depicted in Figure 3 and 4 are not reliant on any correction or comparison with non-introgressed regions, and thus our results regarding sequence similarity and its effect on the repair of DSBs and the amount of genetic shuffling with/without introgression to be novel and important observations for the field.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) Line 149 - this sentence refers to a mixture of papers reporting somatic or meiotic recombination and as these processes are based on different crossover pathways, this should not be mixed. For example, it is known that in *Arabidopsis* MSH2 has a pro-crossover function during meiotic recombination.

Corrected

(2) What is unclear to me is how the crosses are planned. Line 308 shows that there were only two crosses (one "natural" and one "fermentation"), but I understand that this is a shorthand and in fact several (four?) different strains were used for the "fermentation cross". At least that's what I concluded from Fig. 1B and its figure caption. This needs to be further explained. Were different strains used for each fermentation cross, or was one strain repeated in several crosses? In Figure 1, it would be worth showing, next to the panel showing "fermentation cross", a diagram of how "natural cross" was performed, because as I understand it, panel A illustrates the procedure common to both types of crosses, and not for "natural cross".

We thank the reviewer for drawing our attention to confusion about how our crosses were created. We performed two crosses, as depicted in Figure 1A. The fermentation cross is a single cross from two strains isolated from fermentation environments. The natural cross is a single cross from two strains isolated from a tree and insect. Table S1 and the methods section "Strain and library construction" describe the strains used in more detail. We modified Figure 1 and the figure legend to help clarify this. See also response to comment #37.

(3) The authors should provide a more detailed characterization of the genetic differences between chromosomes in their hybrids. What is the level of polymorphism along the *S. uvarum* chromosomes used in the experiments? Is this polymorphism evenly distributed? What are the differences in the level of polymorphism for individual introgressions? Theoretically, this data should be visible in Figure 2D, but this figure is practically illegible in the present form (see next comment).

As suggested, we remade Figure 2D to only include chromosomes with an introgression present, and moved the remaining chromosomes to the supplements (Figure S11). The patterns of markers (which are fixed differences between the strains in the focal cross) should be more clear now. As we detail in the Methods line 507-508, we utilized a total of 24,574 markers for the natural cross and 74,619 markers for the fermentation cross (the higher number in the fermentation cross being due to more fixed differences in regions of introgression).

(4) Figure 2D should be prepared more clearly, I would suggest stretching the chromosomes, otherwise, it is difficult to see what is happening in the introgression regions for CO and NCO (data for SNPs are more readable). Maybe leave only the chromosomes with introgressions and transfer the rest to the supplement?

See previous comment.

(5) How are the Y scales defined for Figure 2D?

Figure 2D now includes units for the y-axis.

(6) Are increases in CO levels in fermentation cross-observed at the border with introgressions? This would indicate local compensation for recombination loss in the

introgressed regions, similar to that often observed for chromosomal inversions.

We see no evidence of an increase in CO levels at the borders of introgressions, neither through visual inspection or by comparing the average CO rate in all fermentation windows to that of windows at the edges of introgressions. This is included in the Discussion lines 360-366, "While we are limited in our interpretations by only comparing two crosses (one cross with heterozygous introgression and one without introgression), these results are in line with findings in inversions, where heterozygotes show sharp decreases in COs, but the presence of NCOs in the inverted region (Crown et al., 2018; Korunes & Noor, 2019). However, unlike heterozygous inversions where an increase in COs is observed on freely recombining chromosomes (the inter-chromosomal effect), we do not see an increase in COs on the borders flanking introgression or on chromosomes without introgression."

(7) Line 336 - "We find positive correlations between CO counts..." - you should indicate here that between fermentation and natural crosses, it was quite hard for me to understand what you calculated.

We corrected the language as suggested.

(8) The term "homology" usually means "having a common evolutionary origin" and does not specify the level of similarity between sequences, thus it cannot be measured. It is used incorrectly throughout the manuscript (also in the intro). I would use the term "similarity" to indicate the degree of similarity between two sequences.

We corrected the language as suggested throughout the document.

(9) Paragraph 360 and Figure 3 - was the "sliding window" overlapping or non-overlapping?

We added clarifying language to the text in both places. We use a 101bp sliding window with 50bp overlaps.

(10) Line 369 - what is "...the proportion of bases that are expected to match between the two parent strains..."?

We clarified the language in this location, and hopefully changes associated with the comment about sequence similarity will make the comment even clearer in context.

(11) Line 378 - should it refer to Figure S1 and not Figure 4?

Corrected.

(12) Line 399 - should refer to Figure 4, not Figure 5.

Corrected

(13) Line 444-449 - the analysis of loss of shuffling in the context of the location of introgression on the chromosome should be presented in the result section.

We shifted the core of the analysis to the results, while leaving a brief summary in the discussion.

(14) The authors should also take into account the presence of indels in their analyses, and they should be marked in the figures, if possible.

We filtered out indels in our variant calling. However, we did analyze our crosses for the presence of large insertions and deletions (Table S2), which can obscure true recombination rates, and found that they were not an issue in our dataset.

Reviewer #2 (Recommendations For The Authors):

This reviewer suggests that the authors address the different points raised in the public review.

(1) This reviewer would like to challenge the relevance of the r -parameter in light of chromosome 12 which has no introgression and still a strong depletion in r in the fermentation cross.

We added this text to the Results, lines 377-381, "While it is unclear what potential mechanism is mediating the difference in shuffling on chromosome 12, we note that the rDNA locus on chromosome 12 is known to differ dramatically in repeat content across strains of *S. cerevisiae* (22–227 copies) (Sharma et al. 2022), and we speculate that differences in rDNA copy number between strains in our crosses could impact shuffling."

(2) This reviewer insists on making sure that NCO detection is unaffected by the marker density, notably in the highly polymorphic regions, to unambiguously support Figure 1C.

We've changed our correction for resolution to be more aggressive (see response to comment #2), and believe we have now adequately adjusted for marker density (see response to comment #38).

Reviewer #3 (Recommendations For The Authors):

I regret using such harsh language in the public review, but in my opinion, there has been a serious error in how marker densities are corrected for, and, since the manuscript is now public, it seems important to make it clear in public that I think that the conclusions of the paper are likely to be incorrect. I regret the distress that the public airing of this may cause. Below are my major concerns:

*(1) The paper is written in a way that makes it difficult to figure out just what the sequence differences are within the crosses. Part of this is, to be frank, the unusual way that the crosses were done, between more than one segregant each from two diploids in both natural and fermentation cases. I gather, from the homology calculations description, that each of these four diploids, while largely homozygous, contained a substantial number of heterozygosities, so individual diploids had different patterns of heterology. Is this correct? And if so, why was this strategy chosen? Why not start with a single diploid where all of the heterologies are known? Why choose to insert this additional complication into the mix? It seems to me that this strategy might have the perverse effect of having the heterology due to the polymorphisms present in one diploid affect (by correction) the impact of a noncrossover that occurs in a diploid that lacks the additional heterology. If polymorphic markers are a small fraction of total markers, then this isn't such a great concern, but I could not find the information anywhere in the manuscript. As a courtesy to the reader, please consider providing at the beginning some basic details about the starting strains-what is the average level of heterology between natural A and natural B, and what fraction of markers are polymorphic; what is the average level of heterology between fermentation A and fermentation B in non-introgressed regions, in introgressed regions, and what fraction of markers are polymorphic? How do these levels of heterology compare to what has been examined before in whole-genome hybrid strains? It also might be worth looking at some of the old literature describing *S. cerevisiae*/*S. carlsbergensis* hybrids.*

We thank the reviewer for drawing our attention to confusion about the cross construction. These crosses were conducted as is typical for yeast genetic crosses: we crossed 2 genetically distinct haploid parents to create a heterozygous diploid, then collected the haploid products of meiosis from the same F1 diploid. Because the crosses were made with haploid parents, it is not possible for other genetic differences to be segregating in the crosses. We have revised Figure 1 and its caption to clarify this. Further details regarding the crosses are in the Methods section "Strain and library construction" and in Supplemental Table S1. We only utilized genetic markers that are fixed differences between our parental strains to call CO and NCO. As we detail in the Methods line 507-508, we utilized a total of 24,574 markers for the natural cross and 74,619 markers for the fermentation cross (the higher number in the fermentation cross being due to more fixed differences in regions of introgression). We additionally revised Figure 2D (and Figure S11) to help readers better visualize differences between the crosses.

(2) There are serious concerns about the methods used to identify noncrossovers and to normalize their levels, which are probably resulting in an artifactually high level of calculated crossovers in Figure 2. As a primary indication of this, it appears in Figure 2 that the total frequency of events (crossovers + noncrossovers) in heterozygous introgressed regions are substantially greater than those in the same region in non-introgressed strains, while just shifting of crossovers to noncrossovers would result in no net increase. The simplest explanation for this is that noncrossovers are being undercounted in non-introgressed relative to introgressed heterozygous regions. There are two possible reasons for this: i. The exclusion of all noncrossover events spanning less than three markers means that many more noncrossovers in introgressed heterozygous regions than in non-introgressed. Assuming that average non-homology is 5% in the former and 1% in the latter, the average 3-marker event will be 60 nt in introgressed regions and 300 nt in non-introgressed regions - so many more noncrossovers will be counted in introgressed regions. A way to check on this - look at the number of crossover-associated markers that undergo gene conversion; use the fraction that involves < 3 markers to adjust noncrossover levels (this is the strategy used by Mancera et al.). ii. The distance used for noncrossover level adjustment (2kb) is considerably greater than the measured average noncrossover lengths in other studies. The effect of using a too-long distance is to differentially under-correct for noncrossovers in non-introgressed regions, while virtually all noncrossovers in heterozygous introgressed regions will be detected. This can be illustrated by simulations that reduce the density of scored markers in heterozygous introgressed regions to the density seen in non-introgressed regions. Because these concerns go to the heart of the conclusions of the paper, they must be addressed quantitatively - if not, the main conclusions of the paper are invalid.

We adjusted the correction factor (See also response to comment #2) and compared the average number of CO and NCO events in introgressed and non-introgressed regions between crosses (two comparisons: introgression CO+NCO in natural cross vs introgression CO+NCO in fermentation cross; non-introgression CO+NCO in natural cross vs non-introgression CO+NCO in fermentation cross). We found no significant differences between the crosses in either of the comparisons. This indicates that the distribution of total events is replicated in both crosses once we correct for resolution.

(3) It is important to distinguish the landscape of double-strand breaks from the landscape of recombination frequencies. Double-strand breaks, as measured by uncalibrated levels of Spo11-linked oligos, is a relative number - not an absolute frequency. So it is possible that two species could have a similar break landscape in terms of topography but have absolute levels higher in one species than in the other.

We agree with this statement, however, we have removed the relevant text to streamline our introduction.

| (4) Lines 123-125. *Just meiosis will produce mosaic genomes in the progeny of the F1; further backcrossing will reduce mosaicism to the level of isolated regions of introgression.*

Adjusted the language to be more specific.

| (5) *Please provide actual units for the Y axes in Figure 2D.*

We have corrected the units on the axes.

| (6) *Tables (general). Are the significance measures corrected for multiple comparisons?*

In Table 3, the cutoff was chosen to be more conservative than a Bonferroni corrected $\alpha=0.01$ with 9 comparisons (0.0011). In text, any result referred to as significant has an associated hypothesis test with a p-value less than its corresponding Bonferroni-corrected α of 0.05. This has been clarified in the caption for Table 3 and in the text where relevant.

<https://doi.org/10.7554/eLife.96184.2.sa0>